

Types of Study

Data Structures: Types of Study

Colin Hannigan

CSB/SJU

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Where We Left Off

Ordinal DV: Ordered logistic regression (Ologit)

Nominal DV: Multinomial logistic regression (Mlogit)

Both models tests how likely *each* outcome is compared to what we'd expect if there were no relationship, relative to the other outcomes.

- H_0 : the β coefficients are 0
- H_1 : the β coefficients are different from 0

The β coefficients (estimates) are not directly interpretable, so we can only speak in terms of likelihood, rather than specific unit changes in the DV (just like binary/binomial logit).

We understand Ologit as predicting the likelihood of being in the **higher categories of the DV**, but Mlogit compares categories independently.

Types of Study

When we think about research design, we should think about what the data will look like.

More specifically, what should the data structure look like (as a table):

ID	Time	w	x	y	z

Cross-Sectional Data

Cross-sectional studies analyze many units during one time period.

Pros:

- Compares units (like countries, people, states) to one another
- Keeps time constant (so no confounding variable due to time period).

Cons:

- Difficult to establish causal direction (“temporal precedence”)
- Introduces other confounding variables unique to each unit (e.g. culture, location, demographics, etc.).

Example: How does **countries’** military spending influence **their** economic development?

Cross-sectional Data Example

#	ID	Time	Spending(\$bn)	GDP/cap	x3	x4
1	USA	2018	649	65,111		
2	China	2018	250	10,098		
3	Saudi	2018	67.6	22,865		
4	India	2018	66.5	2,171		
5	France	2018	63.8	41,760		
6	Russia	2018	61.4	11,162		
7	UK	2018	50	41,030		
8	Germany	2018	49.5	46,563		
...	...	...	...	...	...	...
196	Tuvalu	2018	0.015	3,834		

Time Series Data

Time series (or “longitudinal”) studies analyze one unit across multiple time periods.

Pros:

- Keeps unit constant to avoid other confounding factors that are unique to it.
- Additional time helps establish temporal precedence (for causal direction).

Cons:

- The phenomenon might be unique to that unit!
- We can generalize across time, but not to other units.

Example: How does **the U.S.’s** military spending influence **its** economic development?

Time Series Data Example

#	ID	Time	Spending(\$bn)	GDP/cap	x3	x4
1	USA	2018	649	65,111		
2	USA	2017	605.8	59,927		
3	USA	2016	612.89	59,904		
4	USA	2015	616.48	56,803		
5	USA	2014	631.51	55,032		
6	USA	2013	673.1	53,106		
7	USA	2012	731.09	51,603		
8	USA	2011	775.16	49,883		
...	...	...	...	...	...	...
N	USA	2000	429.45	36,334		

Panel (or “time series cross-sectional”) studies analyze multiple unit across multiple time periods.

Pros:

- Accounts for confounding trends that affects all units at some times (e.g. global recessions, disasters, pandemics, etc.)
- We can generalize over time and to other units!

Cons:

- Tough to find reliable data across some units or for some time periods.
- Makes for a big dataset that can be difficult to manage.

Example: How do **countries'** military expenditures influence **their** economic development **over time**?

Panel Data Example

#	ID	Time	Spending(\$bn)	GDP/cap	x3	x4
1	USA	2018	649	65,111		
2	USA	2017	605.8	59,927		
3	USA	2016	612.89	59,904		
4	China	2018	250	10,098		
5	China	2017	227.83	8,759		
6	China	2016	216.03	8,078		
7	Saudi	2018	67.6	22,865		
8	Saudi	2017	70.4	20,803		
9	Saudi	2016	63.67	19,879		
...	...	...	...	...	...	...

Aggregating/Disaggregating

Sometimes it's useful to break down the units of observation or the time units, even if we have lots of data.

Example 1: Maybe instead of looking at one single observation (the U.S. in 2018) we should look at the state level instead of the country level.

Example 2: Maybe the thing we're examining varies over smaller time units, so we could look at months instead of years.

Pro: it gives us more observations to make sure we can make a better statistical argument (sample becomes "big enough")

Con: There might be different patterns at different levels of aggregation.

Ecological inference problem: (a.k.a. the ecological fallacy) assuming that the pattern in the aggregated units is the same for the disaggregated units.

How to Pick Units and Times

1. Don't be arbitrary (be systematic)
2. Rely on your theory!
3. Data availability
4. Sometimes aggregating/disaggregating is useful

If you want to look at how a Minnesota policy change in 2012 influenced voter turnout, what would the data look like?

Let's think it through...

If you want to look at how a Minnesota policy change in 2012 influenced voter turnout, what would the data look like?

- Could we have time series data?
- Could we have panel data (maybe we disaggregate by county/city)?
- Could we look at a few years leading up to 2012 and a few years after?
- Could we maybe we look at months instead?
- We have to consider when and how often elections are held...

Think about how other variables might influence the time units we look at even more specifically (e.g., Governors' term lengths, legislative sessions, how cold the winters were, if the Packers were winning, etc.).